

Minor Research Project Report

on

Post-Combustion Carbon Capture: A Systematic Aspen Plus Simulation Study

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APPROVAL SHEET

This Project Work entitled “**Post-Combustion Carbon Capture: A Systematic Aspen Plus Simulation Study**” by Mohd. Shayan Khan, K. Bhavani, Indu Priya. G and Sumit Biswas.

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CERTIFICATE

This is to certify that the Project work entitled “**Post-Combustion Carbon Capture: A Systematic Aspen Plus Simulation Study**” is a Bonafide record of work carried out by **Mohd. Shayan Khan, K. Bhavani, Indu Priya. G and Sumit Biswas**, submitted to the faculty of Department of Chemical Engineering, as part of Minor Research Project at National Institute of Technology, Warangal during the academic year 2025-2026.

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ABSTRACT

This study presents a detailed simulation and comparative analysis of post-combustion CO₂ capture using two widely used amine-based solvents: monoethanolamine (MEA) and a blended solvent system of methyl diethanolamine with monoethanolamine (MDEA+MEA). The simulation is based on a real industrial case study of a typical coal-fired power plant, which emits flue gas containing approximately 18 mol% CO₂.

The process is modeled using Aspen Plus, focusing on CO₂ capture through an absorption–desorption system. The primary objective of the study is to achieve at least 90% CO₂ capture efficiency while evaluating the energy requirement and performance of each solvent system.

Key parameters such as solvent flow rate, absorption efficiency, mass transfer characteristics, and reboiler duty are analyzed and compared. The results provide insights into the effectiveness of different solvents in terms of capture efficiency and energy consumption, helping to identify the most suitable solvent system for industrial-scale carbon capture applications.

TABLE OF CONTENTS

Chapter 1: Introduction	1
Chapter 2: Literature Review	3
2.1 Amine-Based Solvents and Their Industrial Relevance	
2.2 Monoethanolamine (MEA)	
2.3 Thermodynamic and Kinetic Considerations in Solvent Selection	
2.4 Modeling Approaches: Equilibrium vs. Rate-Based Simulation	
2.5 Benchmarking of MEA and Advanced Solvent Systems	
2.6 MEA vs. MDEA+MEA Blend: Performance Trade-offs	
2.7 Energy Consumption and Process Optimization	
2.8 Solvent Degradation and Environmental Considerations	
2.9 Research Gap and Relevance to Current Study	
2.10 Objective of the Study	
Chapter 3: Simulation	8
3.1 Introduction	
3.2 Pathway (i): CO ₂ Absorption System (CAS)	
3.3 Pathway (ii): CO ₂ Stripping and Solvent Regeneration	
3.4 Reaction Mechanism of Amine Solvents	
Chapter 4: Results and Discussions	15
4.1 Overview of Simulation Results	
4.2 Performance of MEA System	
4.3 Performance of MEA+MDEA System	
Chapter 5: Conclusion	21
Chapter 6: References	22

Chapter 1: Introduction

1. Introduction

The rapid increase in greenhouse gas (GHG) emissions, particularly carbon dioxide (CO_2), has emerged as a major driver of global climate change. Anthropogenic activities such as fossil fuel combustion, industrial processing, and power generation have significantly elevated atmospheric CO_2 concentrations, leading to severe environmental consequences including global warming, rising sea levels, and extreme weather events. Among these sources, fossil fuel-based power plants contribute a substantial share of global CO_2 emissions, making them a critical target for emission reduction strategies.

In response to these challenges, carbon capture and storage (CCS) technologies have gained considerable attention as effective solutions for mitigating CO_2 emissions. Among the various CCS approaches, post-combustion CO_2 capture is one of the most widely adopted technologies due to its ability to be retrofitted into existing power plants without requiring major modifications to the combustion process. In this technique, CO_2 is separated from flue gas after combustion, typically through chemical absorption using liquid solvents.

Flue gas from coal-fired power plants generally contains approximately 11–13 mol% CO_2 at near-atmospheric pressure, which makes chemical absorption a suitable and efficient method for CO_2 removal. Amine-based solvents are the most commonly used absorbents due to their strong chemical affinity for CO_2 and relatively fast reaction kinetics. Among these, monoethanolamine (MEA), diethanolamine (DEA), methyl diethanolamine (MDEA) and monoethanolamine (MEA) and methyl diethanolamine (MDEA) promoted with piperazine (PZ) have been extensively studied and applied in industrial CO_2 capture processes.

Monoethanolamine (MEA) is widely used as the benchmark solvent due to its fast reaction kinetics and high CO_2 absorption efficiency. However, it suffers from high regeneration energy due to strong carbamate formation.

To address this limitation, blended solvent systems such as MDEA+MEA are employed. Methyldiethanolamine (MDEA), a tertiary amine, offers lower regeneration energy because CO_2 is absorbed via bicarbonate formation rather than stable carbamates, resulting in

weaker bonding and easier stripping. However, pure MDEA exhibits slow reaction kinetics, which limits its standalone performance.

The addition of MEA as an activator enhances the absorption rate of MDEA, providing faster kinetics than pure MDEA while maintaining significantly lower regeneration energy compared to pure MEA. This creates a balanced system that improves overall process efficiency in terms of energy consumption and solvent circulation requirements.

MEA is selected as the activator instead of piperazine (PZ) primarily due to its simpler reaction chemistry and well-established reaction mechanisms. Unlike PZ systems, which involve complex multi-species kinetics and additional ionic reactions, MEA-based blends are easier to define and converge within Aspen Plus rate-based models.

Therefore, a comparative evaluation of these solvents under realistic operating conditions is essential to identify the most efficient and economically viable option for large-scale applications. This study focuses on the simulation and comparative analysis of CO₂ capture using these amine-based solvents. By incorporating actual flue gas conditions into the simulation, the study aims to provide practical insights into solvent performance and process optimization.

Chapter 2: Literature Review

Post-combustion carbon capture (PCC) using chemical absorption with aqueous amine solvents is widely recognized as the most mature and industrially viable technology for CO₂ mitigation from flue gas streams [1], [7], [8]. The process typically employs an absorber–stripper configuration, where CO₂ is absorbed at low temperatures and subsequently desorbed using thermal energy in a regeneration column. Despite its technological readiness, PCC is fundamentally constrained by the “energy–solvent–loss” trilemma, where improvements in capture efficiency often lead to higher regeneration energy or increased solvent degradation and loss [6], [14].

2.1 Amine-Based Solvents and Their Industrial Relevance

Amines remain the benchmark solvents due to their strong chemical affinity for CO₂ and well-established industrial deployment [1], [7], [8]. The dominance of amines arises from their favorable kinetics, high selectivity, and compatibility with dilute flue gas streams. However, no single amine satisfies all performance criteria, which has led to increasing interest in blended systems that combine the advantages of different amine classes [13], [20]. Primary amines such as monoethanolamine (MEA) exhibit rapid reaction kinetics via carbamate formation, while tertiary amines such as methyldiethanolamine (MDEA) offer lower regeneration energy through bicarbonate formation pathways [12], [13].

MEA continues to serve as the baseline solvent for benchmarking studies due to its fast absorption kinetics and high capture efficiency [1], [7]. However, its major limitations include high reboiler duty (typically 3.5–4.2 GJ/ton CO₂), significant oxidative and thermal degradation, and high volatility leading to solvent losses [6], [14]. These drawbacks directly increase operational expenditure and motivate the search for improved solvent systems.

2.2 Monoethanolamine (MEA)

MEA is the most widely studied and commercially used solvent for CO₂ capture and is often considered a benchmark for evaluating other solvents [1], [7], [9]. Its popularity is attributed to its rapid reaction kinetics and high absorption efficiency.

However, MEA suffers from several drawbacks. The regeneration of MEA requires a high

amount of energy due to its strong chemical bonding with CO₂, resulting in a high reboiler duty [14], [15].

Additionally, MEA is highly susceptible to thermal and oxidative degradation, particularly in the presence of oxygen and trace impurities in the flue gas, leading to the formation of degradation products that reduce solvent performance [6], [18]. Furthermore, MEA systems are associated with corrosion problems at elevated temperatures, increasing maintenance costs and affecting equipment lifespan. Solvent losses due to evaporation also contribute to operational inefficiencies [6], [19].

2.3 Thermodynamic and Kinetic Considerations in Solvent Selection

The thermodynamic behavior of amine systems is governed by complex electrolyte interactions involving molecular and ionic species. The Electrolyte Non-Random Two-Liquid (eNRTL) model is widely adopted in Aspen Plus simulations to accurately represent these non-idealities [2], [16]. This framework accounts for both short-range molecular interactions and long-range electrostatic effects, making it essential for modeling CO₂–amine–water systems.

From a reaction mechanism perspective, primary amines like MEA follow the zwitterion mechanism, forming stable carbamates that limit their theoretical loading capacity to 0.5 mol CO₂ per mol amine [9], [12]. In contrast, tertiary amines such as MDEA facilitate CO₂ absorption via bicarbonate formation, allowing a higher theoretical loading of up to 1 mol CO₂ per mol amine but at the cost of slower reaction kinetics [12], [13]. This fundamental trade-off between kinetics and thermodynamics underpins the rationale for using blended solvents [13], [20].

2.4 Modeling Approaches: Equilibrium vs. Rate-Based Simulation

Accurate simulation of PCC processes is critical for solvent benchmarking. Equilibrium-based models (RadFrac) systematically underestimate energy requirements and fail to converge under realistic operating conditions [2], [15]. In contrast, rate-based models (RateSep) incorporate mass transfer resistance and reaction kinetics, providing more realistic predictions of absorber height, temperature profiles, and reboiler duty [2], [17].

Rate-based modeling frameworks implemented in Aspen Plus rely on detailed transport

correlations, film discretization, and kinetic rate expressions. Validated models incorporate electrolyte NRTL thermodynamics, concentration-based kinetics, and experimentally regressed transport properties, ensuring high-fidelity simulation results [16], [17]. These models are essential for capturing performance differences between solvents under identical conditions.

2.5 Benchmarking of MEA and Advanced Solvent Systems

Recent studies emphasize benchmarking solvents under consistent conditions, typically targeting 90% CO₂ capture [4], [5], [14]. MEA remains the reference case, but optimized configurations can reduce its reboiler duty to approximately 3 GJ/ton CO₂ at the expense of increased energy input and potential solvent degradation [5], [14].

Blended solvent systems, particularly those involving tertiary amines, have demonstrated improved performance. MDEA-based systems, when activated with faster-reacting amines, exhibit significantly lower regeneration energy while maintaining acceptable absorption rates [13], [20]. Literature reports indicate that such blends can reduce reboiler duty to 2.8–3.3 GJ/ton CO₂ [4], [15]. Additionally, pilot-scale and industrial studies confirm the effectiveness of such systems in real operating conditions [18], [19].

2.6 MEA vs. MDEA+MEA Blend: Performance Trade-offs

The combination of MDEA with MEA represents a practical compromise between kinetics and energy efficiency. In such blends, MEA provides fast reaction kinetics and high absorption rates, while MDEA contributes lower heat of regeneration and higher loading capacity [13], [20]. This synergistic interaction allows for reduced reboiler duty compared to pure MEA systems while avoiding the kinetic limitations of standalone MDEA.

Literature indicates that blended amines achieve improved cyclic capacity and reduced solvent circulation rates, directly impacting both capital and operational costs [13], [20]. However, these benefits introduce additional complexity in solvent management and reaction modeling.

2.7 Energy Consumption and Process Optimization

The reboiler duty remains the most critical performance indicator for PCC systems, accounting for a major portion of operating costs [14], [15]. This energy requirement

consists of sensible heat, heat of reaction, and latent heat of vaporization.

Optimization strategies focus on minimizing this duty through parameters such as lean loading, liquid-to-gas (L/G) ratio, and stripper pressure [14], [15]. Sensitivity analyses show that reducing lean loading increases absorption driving force but significantly increases regeneration energy, highlighting the need for an optimal operating window or thermodynamic “sweet spot” [14].

2.8 Solvent Degradation and Environmental Considerations

Solvent degradation is a major concern in PCC systems, affecting both performance and environmental compliance [6], [18]. MEA is particularly susceptible to oxidative and thermal degradation, leading to reduced efficiency and increased solvent replacement costs [6].

Increasing MEA concentration may reduce energy consumption but significantly increases solvent loss rates [6], [18]. Blended systems such as MDEA-based solvents show improved resistance to degradation, though challenges such as precipitation and long-term stability remain [18]. Additionally, volatile emissions (“amine slip”) require mitigation strategies such as water wash sections.

2.9 Research Gap and Relevance to Current Study

Despite extensive research, there remains a lack of systematic, simulation-based comparisons under consistent operating conditions, particularly for MEA and MDEA+MEA systems at fixed CO₂ recovery targets [17], [20]. Many studies focus either on energy consumption or capture efficiency independently rather than optimizing all key parameters simultaneously.

Therefore, this study addresses this gap by performing a rigorous Aspen Plus rate-based simulation using the eNRTL framework to benchmark MEA and MDEA+MEA solvents at 90% CO₂ capture. The focus is on minimizing reboiler duty while maintaining acceptable solvent requirements and limiting solvent degradation, thereby addressing the “Energy–Solvent–Loss” trade-off central to PCC system design.

2.10 Objective of the Study

The primary objective of this research is to simulate and compare the performance of selected amine-based solvents for CO₂ capture from flue gas using a realistic industrial case study. The study aims to evaluate the effectiveness of different solvents in achieving high CO₂ capture efficiency while minimizing energy consumption.

The specific objectives of the study are as follows:

- To achieve a target of at least 90% CO₂ capture efficiency under realistic operating conditions.

- To perform a comparative analysis of the following amine-based solvents based on input solvent flowrate, energy consumption and loss of solvent used:
 1. Monoethanolamine (MEA)
 2. Methyl diethanolamine and monoethanolamine (MDEA+MEA)

Chapter 3: Simulation

3.1 Introduction

The present investigation focuses on the simulation and comparative analysis of CO₂ capture from flue gas using amine-based solvents, namely monoethanolamine (MEA) and a blended solvent system of methyl diethanolamine with monoethanolamine (MDEA+MEA). The overall process simulation was carried out using Aspen Plus, employing the ELECNRTL thermodynamic model to accurately represent the vapor–liquid equilibrium and electrolyte behavior of the solvent systems.

The Power Plant operates with a gas mass flow rate of 3039 tons per hour at a temperature of 59.23 °C and a pressure of 1.03 bar.

From the total flue gas stream, a split stream of 18 kmol per hour is considered for further processing in the CO₂ capture unit. Prior to entering the absorber, the flue gas is conditioned to match the operating requirements of the absorption system by reducing its temperature to 313.67 K and adjusting the pressure to 1.01 bar.

3.2 Pathway (i): CO₂ Absorption System (CAS)

The flue gas used in this study was fed to the absorber. The inlet gas stream entered the absorber at an operating temperature of **313.67 K** and a pressure of **1.01 bar**.

The composition of the flue gas is presented in Table 3.1. It mainly consists of nitrogen as the dominant component, followed by carbon dioxide and oxygen, along with minor fractions of water vapor.

Table 3.1: Flue Gas Composition of flowrate = 18 kmol/h

Component	Mole Fraction (%)
N ₂	75.28
CO ₂	18.41
O ₂	4.71
H ₂ O	1.60

The CO₂ capture process was carried out in an adiabatic packed-bed absorber using structured packing (IMTP (Intalox Metal Tower Packing)). The absorber dimensions were kept constant for all solvent systems. The rich solvent leaving the absorber was sent to the stripper for regeneration, where CO₂ was released and the regenerated solvent was excluded.

Table 3.2: Operating Conditions of the CAU

Parameter	MEA	MDEA+MEA
Temperature (K)	313.67	313.67
Pressure (bar)	1.18	1.18
Solvent concentration	30 wt%	MDEA:40wt%; MEA:40 wt%
Solvent flow rate (kmol/h)	222.426	22.51
Absorber diameter (m)	0.427	0.5
Packing height (m)	6.1	6
Packing type	IMTP	BX by SULZER

3.3 Pathway (ii): CO₂ Stripping and Solvent Regeneration

The rich solvent from the absorber is sent to the stripper for regeneration, where the absorbed CO₂ is released through thermal desorption. The process is carried out in a packed column equipped with a reboiler and condenser. Heat supplied in the reboiler facilitates the separation of CO₂ from the solvent, producing a CO₂-rich stream at the top and regenerated solvent at the bottom.

The regenerated solvent is excluded back to the absorber, while the separated CO₂ stream is sent for further processing such as compression and storage. The key operating conditions of the stripper for different solvent systems are summarized in Table 3.3:

Table 3.3 Operating Conditions of Stripper

Parameter	MEA	MDEA+MEA
Pressure (bar)	2.39	2.4
Stripper inner diameter (m)	0.427	0.427
Packing height (m)	6.1	6.1
Packing type	FLEXIPAC 1Y	FLEXIPAC 1Y
Reboiler temperature (K)	365.020302	324.995796
Reboiler heat duty (MW)	250.330474	90.0477967

3.4 Reaction Mechanism of Amine Solvents

The absorption of CO₂ in aqueous amine systems involves a combination of equilibrium and kinetic reactions, including acid–base dissociation, bicarbonate formation, and carbamate formation depending on the solvent type.

The equilibrium constant and reaction rate constant are expressed as functions of temperature using the following relations:

Equilibrium Constant (K_{eq}):

$$\ln K_{eq} = A + \frac{B}{T} + C \ln(T) + D$$

Where:

- K_{eq} = Equilibrium constant
- T = Temperature (K)
- A, B, C, D = Empirical constants

This equation is used for **equilibrium reactions** (like protonation, bicarbonate formation).

Rate Constant (k): Arrhenius Equation

$$k = k_0 \exp\left(\frac{-E}{RT}\right)$$

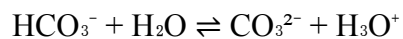
Where:

- k = Reaction rate constant
- k_0 = Pre-exponential factor
- E = Activation energy
- R = Universal gas constant
- T = Temperature (K)

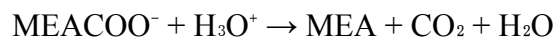
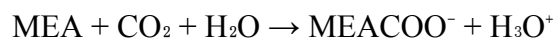
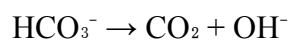
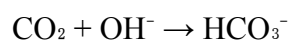
This is used for **kinetic reactions** (like carbamate formation, CO₂ hydration).

(a) MEA (Monoethanolamine)

Equilibrium Reactions:



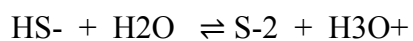
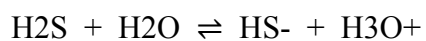
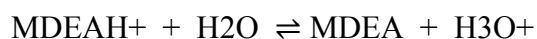
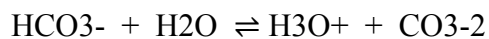
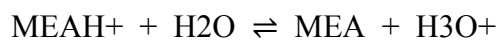
Kinetic Reactions:



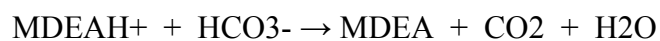
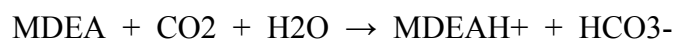
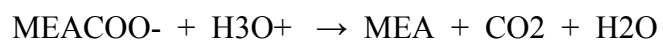
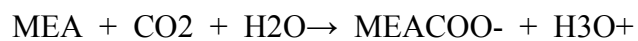
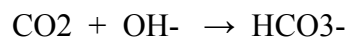
No.	Reaction	k [kmol/m ³ ·s]	n	E (cal/mol)	Phase
4	$\text{CO}_2 + \text{OH}^- \rightarrow \text{HCO}_3^-$	4.32×10^{13}	0	13249	Liquid
5	$\text{HCO}_3^- \rightarrow \text{CO}_2 + \text{OH}^-$	2.38×10^{17}	0	29451	Liquid
6	$\text{MEA} + \text{CO}_2 + \text{H}_2\text{O} \rightarrow \text{MEACOO}^- + \text{H}_3\text{O}^+$	9.77×10^{10}	0	9855.8	Liquid
7	$\text{MEACOO}^- + \text{H}_3\text{O}^+ \rightarrow \text{MEA} + \text{CO}_2 + \text{H}_2\text{O}$	3.23×10^{19}	0	15655	Liquid

(b) MEA+MDEA (Blended Solvent)

Equilibrium Reactions:



Kinetic Reactions:



No.	Reaction	k (kmol/ m ³ ·s)	n	E (cal/mol)	Phase
7	$\text{CO}_2 + \text{OH}^- \rightarrow \text{HCO}_3^-$	4.32×10^{13}	0	13249	Liquid
8	$\text{HCO}_3^- \rightarrow \text{CO}_2 + \text{OH}^-$	2.38×10^{17}	0	29451	Liquid
9	$\text{MEA} + \text{CO}_2 + \text{H}_2\text{O} \rightarrow$ $\text{MEACOO}^- + \text{H}_3\text{O}^+$	9.77×10^{10}	0	9855.8	Liquid
10	$\text{MEACOO}^- + \text{H}_3\text{O}^+ \rightarrow$ $\text{MEA} + \text{CO}_2 + \text{H}_2\text{O}$	3.23×10^{19}	0	15655	Liquid
11	$\text{MDEA} + \text{CO}_2 + \text{H}_2\text{O} \rightarrow$ $\text{MDEAH}^+ + \text{HCO}_3^-$	2.22×10^7	0	9029	Liquid
12	$\text{MDEAH}^+ + \text{HCO}_3^- \rightarrow$ $\text{MDEA} + \text{CO}_2 + \text{H}_2\text{O}$	1.06×10^{16}	0	25424	Liquid

Chapter 4: Results and Discussions

4.1 Overview of Simulation Results

The CO₂ capture process was simulated using Aspen Plus for three amine-based solvent systems: monoethanolamine (MEA) and a blended solvent system of methyl diethanolamine with monoethanolamine (MDEA+MEA). The performance of each solvent was evaluated based on key parameters such as CO₂ capture efficiency, solvent circulation rate, temperature profiles, mass transfer behavior, and energy requirement for regeneration.

A minimum target of **90% CO₂ capture efficiency** was maintained for all solvent systems to ensure a consistent basis for comparison.

4.2 Performance of MEA System

(a) Process Description

The MEA-based system operates under the specified absorber and stripper conditions, where CO₂ is primarily captured through carbamate formation. The process configuration follows the standard absorption–regeneration cycle.

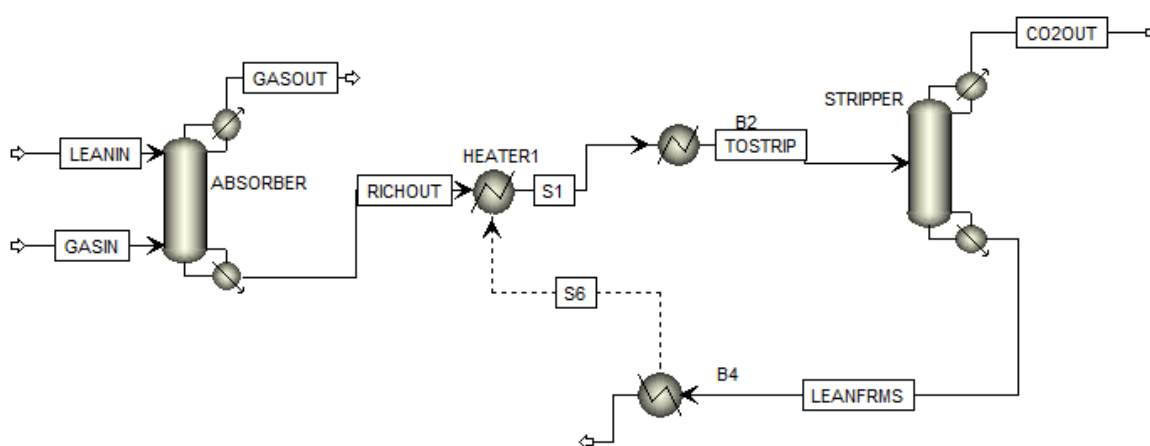


Fig 4.1: MEA-Flowsheet

Table 4.1: Performance of MEA for 90% CO₂ Capture

Parameter	Value	Unit
CO ₂ Capture Efficiency	90	%
Solvent Flow Rate	222.426	KMOL/SEC
Heater Duty	134.967	KW
Reboiler Heat Duty	250.33	KW
Condenser Heat Duty	-139.564	KW
Stripper Pressure	2.394	bar
Stripper Temperature	354.33	K
Reboiler Temperature	365.02	K
Condenser Temperature	297	K

Table 4.2: Absorber Summary

		FLUEGAS	GASOUT	LEANIN	RICHOUT
Temperature	K	332.377	314.038	313.332	327.217
Pressure	Bar	1.0338	1.01325	1.702	1.01
Mole Flows	kmol/hr	18.4	16.228	222.426	221.599
MEA	kmol/hr	0	6.8E-3	12.168	6.15
CO ₂	kmol/hr	3.398	0.339	6.29E-6	4.5E-3

CO₂ Capture Efficiency Equation

$$CO_2 \text{ Capture Efficiency (\%)} = \frac{G_{in} x_{CO_2,in} - G_{out} x_{CO_2,out}}{G_{in} x_{CO_2,in}} \times 100$$

Where:

- G_{in} = Inlet gas flow rate
- G_{out} = Outlet gas flow rate
- $x_{CO_2,in}$ = Mole fraction of CO₂ at inlet
- $x_{CO_2,out}$ = Mole fraction of CO₂ at outlet

CO₂ (inlet) = **3.39 kmol/hr**CO₂ (outlet) = **0.339 kmol/hr**

$$CO_2 \text{ Capture Efficiency} = \frac{3.39 - 0.339}{3.39} \times 100 = 90.0\%$$

Table 4.3: Stripper Summary

		RICH IN	CO2OUT	LEANOUT
Mole Flows	kmol/hr	221.599	2.239	221.501
Temperature	K	354.33	297	365.02
Pressure	Bar	2.394	0.689	0.6936

Table 4.3: Stripper Summary

	RICHIN (kmol/hr)	CO2OUT (kmol/hr)	LEANOUT (kmol/hr)
Mole Flows	221.5995	2.239051	221.501
MEA	6.324097	2.47E-10	10.48739
H2O	193.5245	0.09712	193.5878
CO2	0.001033	2.140931	0.000734
H3O+	3.09E-08	0	2.75E-0
OH-	0.000794	0	0.001138
HCO3-	0.327993	0	0.210358
CO3-2	0.086786	0	0.043601
MEAH+	10.91783	0	8.734356
MEACOO-	10.41547	0	8.435659
N2	0.000898	0.000898	8.21E-18
O2	0.000102	0.000102	2.32E-15
CO	0	0	0
H2	0	0	0
H2S	0	0	0
HS-	0	0	0
S-2	0	0	0

4.3 Performance of MEA+MDEA System

(a) Process Description

The blended MEA+MDEA system enhances CO₂ absorption by combining the high capacity of MDEA with the fast reaction kinetics of MEA.

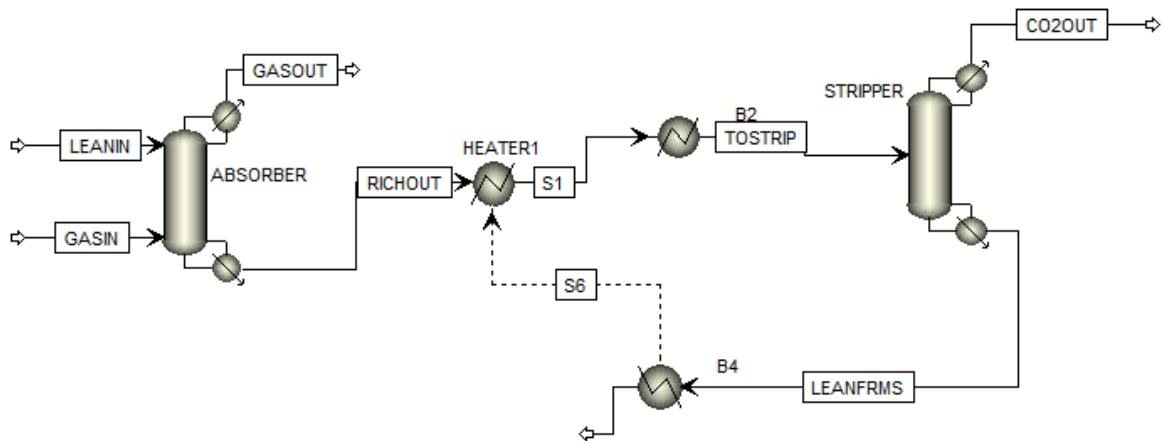


Fig 4.2: MDEA+MEA-Flowsheet

Table 4.4: Performance of MEA+MDEA for 90% CO₂ Capture

Parameter	Value	Unit
CO ₂ Capture Efficiency	90.00	%
Solvent Flow Rate	22.5103	kmol/hr
Heater Duty	25.6544	KW
Reboiler Heat Duty	90.04	KW
Condenser Heat Duty	-57.71	KW
Stripper Pressure	2.4	Bar
Stripper Temperature	398.15	K
Reboiler Temperature	364.996	K
Condenser Temperature	359.147	K

Table 4.5: Absorber Summary

		GASOUT	FLUEGAS	LEANIN	RICHOUT
Temperature	K	297.924	313.67	313.67	301.954
Pressure	bar	1.01325	1.01325	1.01	1.0138
Mole Flows	kmol/hr	2.2164	2.3648	22.513	22.446
CO ₂	kmol/hr	0.0236	0.2364	6.3e-6	4.9e-3

CO₂ Capture Efficiency Equation

$$CO_2 \text{ Capture Efficiency (\%)} = \frac{G_{in} x_{CO_2,in} - G_{out} x_{CO_2,out}}{G_{in} x_{CO_2,in}} \times 100$$

Where:

- G_{in} = Inlet gas flow rate
- G_{out} = Outlet gas flow rate
- $x_{CO_2,in}$ = Mole fraction of CO₂ at inlet
- $x_{CO_2,out}$ = Mole fraction of CO₂ at outlet

CO₂ (inlet) = **0.2364 kmol/hr**CO₂ (outlet) = **0.0236 kmol/hr**

$$CO_2 \text{ Capture Efficiency} = \frac{0.2364 - 0.0236}{0.2364} \times 100 = 90.00\%$$

Table 4.6: Stripper Summary

		TO STRIP	LEANOUT	CO2OUT
Mole Flows	kmol/hr	22.4631	2.239	221.501
Temperature	K	398.15	297	365.02
Pressure	Bar	2.4	0.689	0.6936

Table 4.6: Stripper Summary

Mole Flows	UNITS	TOSTRIP	CO2OUT	LEANFRMS
Total Mole Flow	kmol/sec	0.00624	0.001172	0.005213
MEA	kmol/sec	3.07E-05	1.88E-09	0.000185
H2O	kmol/sec	0.005758	0.001022	0.004787
CO2	kmol/sec	4.79E-06	0.00015	2.42E-09
H3O+	kmol/sec	5.77E-12	0	8.64E-14
OH-	kmol/sec	8.44E-09	0	3.62E-08
HCO3-	kmol/sec	5.11E-05	0	6.30E-07
CO3-2	kmol/sec	8.45E-07	0	1.12E-07
MEAH+	kmol/sec	7.31E-05	0	1.28E-0
MEACOO-	kmol/sec	0.000109	0	1.49E-05
MDEA	kmol/sec	0.000124	1.71E-10	0.000209
MDEAH+	kmol/sec	8.83E-05	0	3.05E-06
N2	kmol/sec	4.12E-08	4.12E-08	7.20E-24
H2S	kmol/sec	0	0	0
HS-	kmol/sec	0	0	0
S-2	kmol/sec	0	0	0
O2	kmol/sec	0	0	0
CO	kmol/sec	0	0	0

Chapter 5: Conclusion

The study concludes that the blended solvent system of MEA + MDEA significantly outperforms pure MEA in both absorber and stripper performance under identical CO₂ capture conditions (~90%). In the absorber, the blended system achieves much higher efficiency with a drastically reduced solvent flow rate (from ~222.43 kmol/hr for MEA to ~22.51 kmol/hr), owing to the synergistic effect of fast reaction kinetics from MEA and high CO₂ loading capacity from MDEA. This results in lower operational requirements and improved mass transfer. In the stripper, the MEA + MDEA system demonstrates substantial energy savings, with reboiler heat duty reduced from ~250 kW to ~90 kW, along with lower condenser duty, indicating reduced thermal demand. The blend also shows lower solvent consumption per unit CO₂ captured, reduced solvent loss, and significantly lower specific energy requirement (~715 to ~180 MJ/kmol CO₂). Despite operating at slightly higher temperatures in the stripper, the system enhances regeneration efficiency without increasing energy input.

Overall, the MEA + MDEA blend provides a balanced and optimized solution by overcoming the high energy penalty of MEA while maintaining effective absorption performance, making it a more energy-efficient and industrially viable option for post-combustion CO₂ capture.

Table 5.1: Conclusion Table

Parameter	MEA	MEA+MDEA
Solvent flow rate (kmol/hr)	222.426	22.5103
Reboiler temperature (K)	365.020302	324.995796
Reboiler heat duty (KW)	250.330474	90.0477967
Amount of solvent input/kmol of CO ₂ captured	~176.88 kmol/kmol _{CO₂}	~105.77 kmol/kmol _{CO₂}
Amount of solvent Loss /kmol of CO ₂ captured	~0.0005	~0.00005
Reboiler Duty(MJ)/kmol of CO ₂ captured	~715 MJ/kmol _{CO₂}	~180 MJ/kmol _{CO₂}

Chapter 6: References

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